

Name: Sanjana Guttapally

SRN: PES1UG24CS420

Section: G

Lab 3 - Architectural Justification: Virtual Technical Conference Platform

Architecture Selection: I chose Microservices Architecture for the Virtual Technical Conference Platform.

Reason 1: Very Different Scaling Needs Across Functions

The platform covers paper review, reviewer assignment, schedule generation, and live Q&A, and these have completely different load patterns. Live Q&A only sees heavy traffic while a session is actually running and needs to support many attendees posting and upvoting questions at once, while paper review and reviewer assignment mostly see activity during the submission and review window, and scheduling only runs when a Track Chair generates or adjusts a timetable.

Microservices lets each of these scale independently instead of forcing the whole platform to scale together for a spike that only affects one function.

Reason 2: Independent Deployment and Fault Isolation

Paper review, reviewer assignment, scheduling, and live Q&A are separate concerns that can be built, updated, and deployed on their own schedules. If the Live Q&A Service needs a fix or update in the middle of a conference, it can be redeployed without touching the Paper Review Service or Scheduling Service, and if it happens to fail during a live session, Track Chairs can still generate schedules and reviewers can still submit their evaluations since those services keep running independently.

Security Advantage

Author identity verification and role based access are handled entirely inside the Identity & Access Control Component, and every other service must call through its IdentityVerificationInterface rather than checking roles on its own. This keeps the double blind review rule enforced in one isolated, focused service instead of being repeated across the Paper Review, Reviewer Assignment, and Scheduling services, which reduces the chance of an author's identity leaking through a service that was not built with that rule in mind.

Performance Benefit

Because the Scheduling Service is its own independent component, it can be given dedicated resources and optimized purely for solving room and speaker conflicts quickly, without competing for processing power with the Paper Review Service or the Live Q&A Service running at the same time. This isolation makes it realistic to consistently meet the requirement of resolving scheduling constraints in under 10 seconds, even while other parts of the platform are under load.

UML Diagram

